jayjay’s Playlists

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Capstone Lightning Talk #2: Spotify Mood Clustering

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DSB-2

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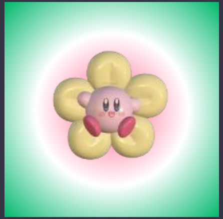
The Problem



Why can’t we just use genre as “mood”?

Because genre is a weak proxy:

- A single genre spans calm and intense tracks alike
- The real mood signal lives in the audio features, not the genre tag
- So clustering has to run on audio features alone – unsupervised

Success = statistically separated AND interpretable clusters, validated on real tracks

Potential Audience

Listeners

Wants to match songs to a mood

Dev Teams

Builds mood discovery features

Analysts

Tracks mood trends in a library

Goals & Objectives

Type

Method

Unsupervised clustering – no target variable

Models

Count

K-Means · DBSCAN · Agglomerative Clustering

Metrics

Checks

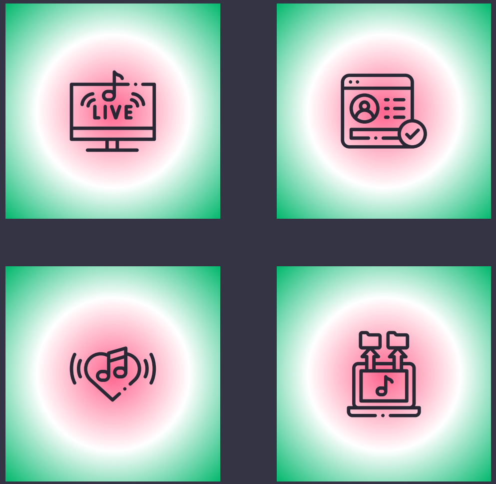
Silhouette, Davies-Bouldin, Elbow, real-track checks

Mood-DNA Feature Dictionary

7 features

#	Feature	Range	Role	Notes
1	energy	0.0-1.0	Core axis	Intensity
2	valence	0.0-1.0	Core axis	Happy vs. sad
3	danceability	0.0-1.0	Texture	Rhythm
4	acousticness	0.0-1.0	Texture	Acoustic
5	loudness	-46 to +1 dB	Intensity	Scale first
6	tempo	0-239 BPM	Intensity	Scale first
7	mode	0 or 1	Weak signal	Major/min or

Data Readiness



What does the data need before clustering?

Three things stood out:

- loudness and tempo sit on totally different scales than the 0-1 features
- energy and loudness correlate at ~0.68 the strongest pair
- Real outliers found in acousticness, loudness, and tempo

So every feature is scaled with RobustScaler first

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Spotify Mood Clustering

Genre ≠ Mood

What We Found

Does genre actually predict mood?

The evidence says no:

- Valence & energy boxplots span a wide range inside every genre
- A valence-vs-energy scatter shows genres overlapping almost completely
- No genre owns a distinct region of the mood plane

Conclusion: mood must be learned from audio features, not inherited from genre

3 Algorithms Compared

K-Means

Primary

The only one here with .predict() for new tracks

DBSCAN

Compared

Checks for natural density structure and outliers

Agglom.

Baseline

Hierarchical clustering – baseline for comparison

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Spotify Mood Clustering

K = 4

Chosen over the silhouette-optimal K=2 (0.285) –
K=4 scores 0.160 but maps to 4 usable moods

Mood Cluster Composition (K=4)

Hype 34%

Highest energy, highest valence

Sunny 21%

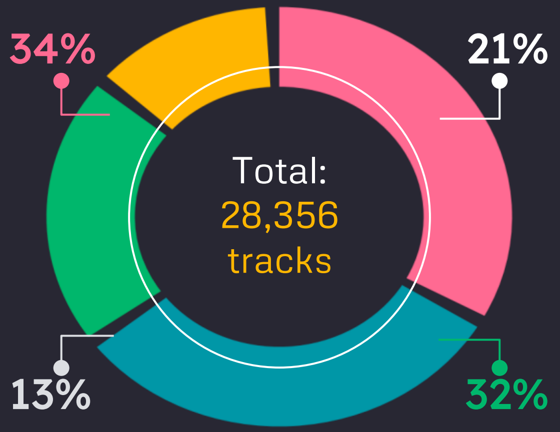
Low energy, high valence

Mellow 13%

Low energy, low valence, high acousticness

Dark 32%

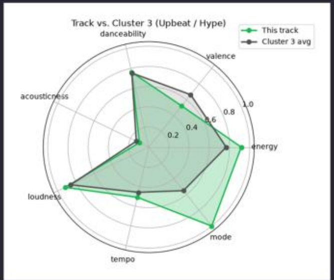
Highest energy, lowest valence



K-Means on the 7 mood-DNA features, validated against real tracks. **See notebook for full breakdown**

LIVE DEMO — SINGLE TRACK

Track vs. Cluster



PREDICTED MOOD

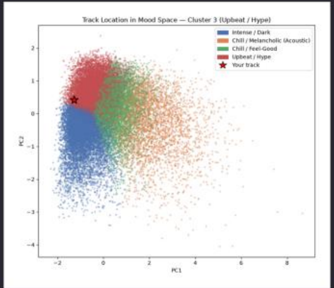
Upbeat / Hype

Here’s the live demo

Compares the track’s audio profile against its cluster average – closer shapes mean a stronger mood match.

LIVE DEMO — SINGLE TRACK

Location in Mood Space



CLUSTER ASSIGNMENT

Cluster 3 (Upbeat / Hype)

Here’s the live demo

The same track in 2D – it sits right on the boundary between Upbeat/Hype and Intense/Dark.

Mood-Matched Playlist (Cluster 3)

#	Track	Artist	Mood	Distance
1	Do It Again	Steve Aoki	Upbeat / Hype	0.324
2	Lettin Go!	Janelle Monáe	Upbeat / Hype	0.328
3	Blah Blah Blah	Kesha	Upbeat / Hype	0.351
4	You Got That	Loud About Us	Upbeat / Hype	0.362
5	Where Is the Love	Alex Martura	Upbeat / Hype	0.392
6	As If It's Your Last	BLACKPINK	Upbeat / Hype	0.393
7	Don't Stop	Curbi	Upbeat / Hype	0.418

Progress: Challenges & Next Steps

Bug Fix

Fixed

Fixed thresholds collapsed 3/4 clusters into one label

Pick K

Resolved

K=2 was statistically better but too coarse – chose K=4

Export

Next

Export the scaler, K-Means model, and PCA as .joblib

Backend

Next

Wrap the pipeline in a class for a Streamlit/Gradio demo

Spotify Mood Clustering

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Thanks!